

No.	Time	Source	Destination	Protocol	Length	Info
132	8.401921	192.168.0.13	103.230.106.2	HTTP	539	GET /mbbs/ HTTP/1.1
134	8.408954	103.230.106.2	192.168.0.13	HTTP	926	HTTP/1.1 200 OK (text/html)
137	8.514317	192.168.0.13	103.230.106.2	HTTP	564	GET /mbbs/top.php HTTP/1.1
141	8.520122	192.168.0.13	103.230.106.2	HTTP	565	GET /mbbs/main.php HTTP/1.1
143	8.522205	103.230.106.2	192.168.0.13	HTTP	628	HTTP/1.1 200 OK (text/html)
145	8.528018	103.230.106.2	192.168.0.13	TCP	10	[TCP Previous segment not captured] 80 → 60661 [PSH, ACK] Seq=436 Ack=512 Win=4591 Len=1025Continuation
149	8.530127	103.230.106.2	192.168.0.13	HTTP	123	[TCP Previous segment not captured] Continuation
151	8.532401	103.230.106.2	192.168.0.13	HTTP	14	[TCP Out-Of-Order] Continuation
45	1021.967	192.168.0.13	199.232.214.1	HTTP	342	GET /msdownload/update/v3/static/trustedr/en/disallowedcertstl.cab?b99f4498167a3adb HTTP/1.1
45	1022.016	199.232.214.1	192.168.0.13	HTTP	257	HTTP/1.1 304 Not Modified

HTTP Request:

Physical (Frame)

```

Wireshark - Packet 132 - Ethernet 2
  Frame 132: Packet, 539 bytes on wire (4312 bits), 539 bytes captured (4312 bits) on interface \Device\NPF_{7B0971C2-3656-4183-819F-7141859279A7}, id 0
    Section number: 1
    Interface id: 0 (\Device\NPF_{7B0971C2-3656-4183-819F-7141859279A7})
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov 11, 2025 22:38:21.471552000 Bangladesh Standard Time
    UTC Arrival Time: Nov 11, 2025 16:38:21.471552000 UTC
    Epoch Arrival Time: 1762879101.471552000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 491.000 microseconds]
    [Time since reference or first frame: 8.401921000 seconds]
    Frame Number: 132
    Frame Length: 539 bytes (4312 bits)
    Capture Length: 539 bytes (4312 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
  
```

This layer handles the packet's physical characteristics, showing it is 539 bytes long and contains protocols: eth:ethertype:ip:tcp:http.

Data Link (Ethernet)

```

Ethernet II, Src: GigaByteTech_86:b4:73 (18:c0:4d:86:b4:73), Dst: DLinkInterna_5f:ee:d2 (c4:e9:0a:5f:ee:d2)
  Destination: DLinkInterna_5f:ee:d2 (c4:e9:0a:5f:ee:d2)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Source: GigaByteTech_86:b4:73 (18:c0:4d:86:b4:73)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 1]
  
```

The frame uses the client's Source MAC address (GigaByteTech) and the router's Destination MAC address (DLinkInterna) to transfer data across the local network segment.

Network (IPv4)

```
▾ Internet Protocol Version 4, Src: 192.168.0.13, Dst: 103.230.106.211
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 525
    Identification: 0xfb6d (64365)
  ▸ 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.13
    Destination Address: 103.230.106.211
    [Stream index: 11]
```

The client's Source IP (192.168.0.13) is used to route the packet to the server's Destination IP (103.230.106.211) across networks.

Transport (TCP)

```
▾ Transmission Control Protocol, Src Port: 60660, Dst Port: 80, Seq: 1, Ack: 1, Len: 485
  Source Port: 60660
  Destination Port: 80
  [Stream index: 4]
  [Stream Packet Number: 4]
  ▸ [Conversation completeness: Complete, WITH_DATA (63)]
    [TCP Segment Len: 485]
    Sequence Number: 1 (relative sequence number)
    Sequence Number (raw): 3537114667
    [Next Sequence Number: 486 (relative sequence number)]
    Acknowledgment Number: 1 (relative ack number)
    Acknowledgment number (raw): 1305548766
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x018 (PSH, ACK)
    Window: 64240
    [Calculated window size: 64240]
    [Window size scaling factor: -2 (no window scaling used)]
    Checksum: 0x956e [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ [Timestamps]
  ▸ [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (485 bytes)
```

The packet uses the client's ephemeral Source Port (60660) and the standard web Destination Port (80) to ensure a reliable connection via the PSH, ACK flags.

Application (HTTP)

```
▼ Hypertext Transfer Protocol
  ▼ GET /mbbs/ HTTP/1.1\r\n
    Request Method: GET
    Request URI: /mbbs/
    Request Version: HTTP/1.1
    Host: dgme.teletalk.com.bd\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8\r\n
    Sec-GPC: 1\r\n
    Accept-Language: en-GB,en;q=0.9\r\n
    Accept-Encoding: gzip, deflate\r\n
    Cookie: PHPSESSID=gpfv4u4t93bsdj59q4860gtfj5\r\n
    Cookie pair: PHPSESSID=gpfv4u4t93bsdj59q4860gtfj5
  \r\n
  [Response in frame: 134]
  [Full request URI: http://dgme.teletalk.com.bd/mbbs/]
```

The client uses the GET method to request the resource /mbbs/ from the server.

HTTP Response:

Physical (Frame)

```
Wireshark - Packet 134 - Ethernet 2
▼ Frame 134: Packet, 926 bytes on wire (7408 bits), 926 bytes captured (7408 bits) on interface \Device\NPF_{7B0971C2-3656-4183-819F-7141859279A7}, id 0
  Section number: 1
  ▼ Interface id: 0 (\Device\NPF_{7B0971C2-3656-4183-819F-7141859279A7})
    Interface name: \Device\NPF_{7B0971C2-3656-4183-819F-7141859279A7}
    Interface description: Ethernet 2
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov 11, 2025 22:38:21.478585000 Bangladesh Standard Time
    UTC Arrival Time: Nov 11, 2025 16:38:21.478585000 UTC
    Epoch Arrival Time: 1762879101.478585000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 3.482000 milliseconds]
    [Time delta from previous displayed frame: 7.033000 milliseconds]
    [Time since reference or first frame: 8.408954000 seconds]
    Frame Number: 134
    Frame Length: 926 bytes (7408 bits)
    Capture Length: 926 bytes (7408 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
```

The frame is 926 bytes long and includes the same protocol stack, confirming the physical transmission of the server's response.

Data Link (Ethernet)

```
▼ Ethernet II, Src: DLinkInterna_5f:ee:d2 (c4:e9:0a:5f:ee:d2), Dst: GigaByteTech_86:b4:73 (18:c0:4d:86:b4:73)
  ▼ Destination: GigaByteTech_86:b4:73 (18:c0:4d:86:b4:73)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ...0 .... = IG bit: Individual address (unicast)
  ▼ Source: DLinkInterna_5f:ee:d2 (c4:e9:0a:5f:ee:d2)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ...0 .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 1]
```

The Source MAC address (DLinkInterna/Router) and the Destination MAC address (GigaByteTech/Client) are reversed to deliver the frame to the client's physical device.

Network (IPv4)

```
▼ Internet Protocol Version 4, Src: 103.230.106.211, Dst: 192.168.0.13
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 912
  Identification: 0xfc60 (64608)
  ▼ 010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 246
  Protocol: TCP (6)
  Header Checksum: 0xf197 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 103.230.106.211
  Destination Address: 192.168.0.13
  [Stream index: 11]
```

The Source IP (103.230.106.211) and Destination IP (192.168.0.13) are reversed, routing the packet back to the correct client.

Transport (TCP)

```
Transmission Control Protocol, Src Port: 80, Dst Port: 60660, Seq: 1, Ack: 486, Len: 872
  Source Port: 80
  Destination Port: 60660
  [Stream index: 4]
  [Stream Packet Number: 6]
  [Conversation completeness: Complete, WITH_DATA (63)]
  [TCP Segment Len: 872]
  Sequence Number: 1      (relative sequence number)
  Sequence Number (raw): 1305548766
  [Next Sequence Number: 873      (relative sequence number)]
  Acknowledgment Number: 486      (relative ack number)
  Acknowledgment number (raw): 3537115152
  0101 .... = Header Length: 20 bytes (5)
  Flags: 0x018 (PSH, ACK)
  Window: 4565
  [Calculated window size: 4565]
  [Window size scaling factor: -2 (no window scaling used)]
  Checksum: 0x8a06 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  [Timestamps]
  [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (872 bytes)
```

The Source Port (80) and Destination Port (60660) are reversed. It uses the Acknowledgment Number (486) to confirm receipt of the client's data and carries the payload.

Application (HTTP)

```
▼ Hypertext Transfer Protocol, has 2 chunks (including last chunk)
  ▼ HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Server: nginx\r\n
    Date: Tue, 11 Nov 2025 16:33:31 GMT\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    Transfer-Encoding: chunked\r\n
    Connection: keep-alive\r\n
    Vary: Accept-Encoding\r\n
    X-Powered-By: PHP/5.6.40\r\n
    X-Frame-Options: SAMEORIGIN\r\n
    X-XSS-Protection: 1; mode=block\r\n
    Content-Encoding: gzip\r\n
    \r\n
    [Request in frame: 132]
    [Time since request: 7.033000 milliseconds]
    [Request URI: /mbbs/]
    [Full request URI: http://dgme.teletalk.com.bd/mbbs/]
  ▶ HTTP chunked response
    Content-encoded entity body (gzip): 562 bytes -> 985 bytes
    File Data: 985 bytes
```

The server sends a response with Status Code 200 (OK), indicating the request was successful and including the HTML content.